

- Q.31 Explain the overall heat transfer coefficient.
 Q.32 Explain the principle & working of ball mill grinder.
 Q.33 Define humidity & explain relative humidity.
 Q.34 Write the difference between crushing and grinding.
 Q.35 State Kick's law and bonds in crushing.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain with a neat sketch, working, uses of the Shell and tube heat exchanger?
 Q.37 Explain construction and working of different types of screen used for separation.
 Q.38 Explain First law and Second law of thermodynamics.

No. of Printed Pages : 4

Roll No.

182241/122241/

032241/095156

4th Semester/ Plastic Subject : Fundamentals of Chemical Engineering / Unit Operation

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a thermodynamics law?
 a) Zeroth law of thermodynamics
 b) Faraday's Law of thermodynamics
 c) Ideal Gas Law of thermodynamics
 d) Boyle's Law of thermodynamics
- Q.2 Heat transfer takes place in liquids and gases is essentially due to
 a) Conduction b) Convection
 c) Radiation d) All of these
- Q.3 Perfect insulator conduct heat and electricity
 a) True b) False
- Q.4 Heat transfer takes place according to which law?
 a) Newton's law of cooling
 b) Second law of thermodynamics
 c) Newton's second law of motion
 d) First law of thermodynamics

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182241/122241/
 032241/095156

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182241/122241/
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- Q.5 The efficiency of the ideal screens is _____.
a) Unity b) Infinity
c) Zero d) Can't be determined
- Q.6 The heat is transferred by conduction, convection and radiation simultaneously in
a) Melting of ice
b) Boiler furnaces
c) Condensation of steam in condenser
d) None of these
- Q.7 Which of the following is a type of thermodynamic system?
a) Open system
b) Closed system
c) Thermally isolated system
d) All of the mentioned
- Q.8 Which of the following is a type of coarse crusher?
a) Rod mill b) Stamp battery
c) Jaw crusher d) Dicer
- Q.9 Which of the following is a method of heat transfer?
a) Convection b) Radiation
c) Conduction d) All of the mentioned
- Q.10 Joule was the first to prove that heat is a type of energy, laying the groundwork for the fundamental law of thermodynamics.
a) False b) True

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Define mass transfer.
- Q.12 Define free convection.
- Q.13 Define crushing.
- Q.14 Define heat.
- Q.15 Define Fick's law.
- Q.16 Define insulator.
- Q.17 Define the Carnot cycle.
- Q.18 Define isothermal process.
- Q.19 Define humidity.
- Q.20 Define mesh number.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the first law of thermodynamics.
- Q.22 Explain Fourier law of heat conduction.
- Q.23 Define the principle of roll mill and give its use.
- Q.24 Define conduction and convection with suitable examples.
- Q.25 Write a short note on Jaw Crusher.
- Q.26 Explain open and closed system with example.
- Q.27 Define Humid Volume and Humid heat.
- Q.28 Draw the sketch of cycle separator explain its working.
- Q.29 Write a short note on Blake Jaw crusher.
- Q.30 Explain Isothermal process of thermodynamics.